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United States Patent	12403947
Kind Code	B1
Date of Patent	September 02, 2025
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Wheel attachment for tamper

Abstract

A wheel attachment for tamper including a wheeled axle assembly with a handle assembly and tamper assembly. The bottom portion of a tamper assembly is mounted in the wheeled axle assembly. The wheeled axle assembly includes a pair of wheels with a bearing portion attached at the endings of the axle with a couple bearings. The axle includes a fastener base which is located at a center of the axle with a differential opposite distance from the handle and allows to set the tamper therein. The handle assembly include a fastener located in an opposite side from the handle allowing to drive the wheeled axle assembly. The base of the tamper can be inserted through the fastener base and be mounted on the axle.

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Family ID:	96882131
Appl. No.:	17/973716
Filed:	October 26, 2022

Publication Classification

Int. Cl.: **B62B5/00** (20060101); **B62B1/10** (20060101); **B62B5/06** (20060101); E01C19/35 (20060101); E02D3/046 (20060101)

U.S. Cl.:

CPC **B62B5/0083** (20130101); **B62B1/10** (20130101); **B62B5/06** (20130101); B62B2202/48 (20130101); E01C19/35 (20130101); E02D3/046 (20130101)

Field of Classification Search

CPC: B62B (5/0083); B62B (1/10); B62B (5/06); B62B (2202/48); E01C (19/35); E02D (3/046)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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8172240	12/2011	Zimmerman	280/47.131	E02D 3/074
9045154	12/2014	Avery	N/A	B62B 5/0083
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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a wheel attachment for tamper and, more particularly, to a wheel attachment for tamper that includes a mobility detachable device with a shaped attachment handle for a tamper to place in.

2. Description of the Related Art

(2) Several designs for a wheel attachment for tamper have been designed in the past. None of them, however, include a detachable dolly wherein a jumping jack can be mounted into a base and can be used as a mobility device to transport over multiple surfaces.

(3) Applicant believes that a related reference corresponds to U.S. Pat. No. 8,128,105 issued for a wheel assembly for providing wheeled transport to a compaction tool having a foot structure. Applicant believes that another related reference corresponds to U.S. Pat. No. 10,300,936 issued for a tamper cart system. However, it differs from the present invention because it includes a mobility attachment device for a tamper comprising an axle with wheels on either end, or a pair of attachment arms for removably attaching the mobility device to the body of a tamper.

(4) Other documents describing the closest subject matter provide for a number of more or less complicated features that fail to solve the problem in an efficient and economical way. None of these patents suggest the novel features of the present invention.

SUMMARY OF THE INVENTION

(5) It is one of the objects of the present invention to provide a wheel attachment for tamper that includes a wheeled axis to easily move over multiple surfaces.

(6) It is another object of this invention to provide wheel attachment for tamper that includes a handle which can be used as a direction device.

(7) It is still another object of the present invention to provide wheel attachment for tamper that includes an axle to mount a tamper device.

(8) It is yet another object of this invention to provide such a device that is inexpensive to implement and maintain while retaining its effectiveness.

(9) Further objects of the invention will be brought out in the following part of the specification,

wherein detailed description is for the purpose of fully disclosing the invention without placing limitations thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) With the above and other related objects in view, the invention consists in the details of construction and combination of parts as will be more fully understood from the following description, when read in conjunction with the accompanying drawings in which:

(2) FIG. 1 represents an operational view wherein a user is moving a tamper assembly **60** that is mounted to the present invention **10**.

(3) FIG. 2 shows an enlarged perspective view of the handle assembly **40**. A handle **42** is attached to wheeled axle assembly **20**. Bearing portions **24a** are attached to wheels **26**.

(4) FIG. 3 illustrates a front isometric view of the wheeled axle **22**. One of the bearings **24** is attached in at least one of the tires **28** which is placed in each of the wheels **26**.

(5) FIG. 4 is a representation of a right perspective view of the wheeled axle assembly **20**. Fastener base **44** is attached to wheeled axle **22**.

(6) FIG. 5 shows an isometric view of the tamper **62** which is mounted to the axle **22**. Bottom portion **64** is inserted through the fastener base **44**.

DETAILED DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

(7) Referring now to the drawings, where the present invention is generally referred to with numeral **10**, it can be observed that it basically includes a wheeled axle assembly **20**, a handle assembly **40** and a tamper assembly **60**. It should be understood there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

(8) Wheeled axle assembly **20** includes a wheeled axle **22**, bearings **24**, a bearing portions **24a** wheels **26** and tires **28**. In a suitable embodiment wheeled axle **22** may have a half box section body as best observed in FIG. 2 with two thin plate sheets perpendicularly connected through the sides of each of plate sheets that conform the wheeled axle **22**. Nevertheless, another other body like tube, pool, hexagon prism, rectangular prism, rod body, pentagonal prism, or any other variation thereof may be suitable to the wheeled axle **22**. In a suitable embodiment, the wheeled axle **22** may be suitable to be made of a steel material. Nevertheless, the wheeled axle **22** may be made of a carbon steel material, alloy steel material, metal material, rigid plastic material, or any other variation thereof. In one embodiment the wheeled axle **22** may be a support for tamper assembly **60**. In another embodiment the wheeled axle **22** may be a support for handle assembly **40**. Both of the bearings **24** may be attached to each of the endings of the wheeled axle **22** through the internal body of bearing **24**. In a suitable embodiment, the bearings **24** may have a circular shape. In a preferred embodiment, bearings **24** may be a type of roller bearing G. Nevertheless, it is to be considered that any other type of bearing wheels like plain bore bearing, ball bearing, central ball bearing, two central ball bearings, spherical roller bearing may be suitable for the bearings **24**. It is to be considered that the bearings **24** may also be attached to each of the wheels **26**. The bearings **24** may be made of a carbon steel. Nevertheless, in other embodiments, the bearings **24** may be made of a stainless steel, carbon steel, tool steel, or any other variation thereof. It is to be understood that the bearings **24** may allow the wheeled axle **22** to roll. The bearing portions **24a** may have a circular body as the bearings **24**. In a suitable embodiment, the bearing portions **24a** may be made of a titanium material to withstand heavy force applied thereof. Nevertheless, it should be considered that other materials like steel, carbon fiber, metal, aluminum alloy, casting

aluminum or any other variation thereof. In a suitable embodiment, the bearing portions **24a** may be a set location for the bearings **24**.

(9) It should be considered that bearing portions **24a** may be concentrically attached to the wheels **26**. The bearings **24** may be concentrically attached to each one of the bearing portions **24a** to attach the wheeled axle **22** to the wheels **26** and the tires **28**. In a preferred embodiment the wheels **26** may be attached to the wheeled axle **22** through the bearings **24**. The wheels **26** may be made of a cast aluminum alloy wheel material to withstand heavy force applied thereof. Nevertheless, it should be considered that other materials like steel wheels, carbon fiber wheels, metal wheels, steel ring, aluminum alloy wheels, casting aluminum or any other variation thereof. The wheels **26** may allow to transport the tamper assembly **60**. Both wheels **26** may be attached to each of the sides of the wheeled axle **22**. The tires **28** may be mounted to each of the wheels **26**. The tires **28** may be made of a synthetic rubber. Nevertheless, in other embodiments the wheels **26** may be made of a butadiene rubber, styrene butadiene rubber material, or any other material thereof. In one embodiment the tires **28** may be a type of all-season tire. Nevertheless, it is to be considered that other different types of tires like touring tires, summer tires, performance tires, highway tires, all-terrain tires, rib tires, spare tires, or any other variation thereof may be suitable for tires **28**.

(10) The handle assembly **40** includes a handle **42** and a fastener base **44**. It is to be considered that handle **42** may be perpendicularly attached to the wheeled axle **22**. In a suitable embodiment, the handle **42** may have a rectangular body as best observed in FIG. 2. Nevertheless, other shapes like linear body, cylindric body, curly body, or any other variation thereof. In a suitable embodiment the handle **42** may have a rectangular body defining a hollow rectangle which extends from the wheeled axle **22**. Nevertheless, in another embodiment, the handle **42** may have a s-shaped body. In a suitable embodiment, the handle **42** may be a fastener for tamper assembly **60**. The handle **42** may be made of a stainless-steel material to withstand when applying any force thereof. Nevertheless, it should be considered that any other material like carbon steel, aluminum alloy, carbon fiber, metal, aluminum, steel, or any other variation thereof, may be suitable for the handle **42**. In a preferred embodiment, the handle **42** may be attached on a half side of the wheeled axle **22**. Nevertheless, the handle **42** may be attached to a top side, bottom side of the wheeled axle **22**.

(11) As best shown in FIG. 2, the handle **42** may be attached on a middle section of the wheeled axle **22** parallelly placed with a difference distance between each other. The handle **42** may be configured to drive the tamper assembly **60** by user's feet. In a suitable embodiment, the handle **42** may be attached to the wheeled axle **22** by a welding technique. Nevertheless, in another embodiment, the handle **42** may be attached to the wheeled axle **22** by a riveting technique. The fastener base **44** may be attached opposite to handle **42** wherein fastener base **44** may be perpendicularly attached on wheeled axle **22** referring to handle **42**. The fastener base **44** may be made of a steel material. Nevertheless, another material like carbon steel material, alloy steel material, metal material, rigid plastic material, copper material, titanium material or any other variation thereof, may be suitable for the fastener base **44**. In one embodiment the fastener base **44** may be a support for the tamper assembly **60**. The fastener base **44** may have a hollow rectangular body as best observed in FIG. 4. In other embodiments, the fastener base **44** may have a rectangular base body, triangular base body, or any other variation thereof. The fastener base **44** may be attached to wheeled axle **22** by a welding technique. Nevertheless, in another embodiment a riveting technique may be suitable for the fastener base **44**. As best shown in FIG. 1 the fastener base **44** may be attached to a bottom side of the wheeled axle **22**. In a suitable embodiment the fastener base **44** may be a support for the tamper assembly **60**. In a suitable embodiment, the fastener base **44** may be attached on a middle portion of the wheeled axle **22**.

(12) The tamper assembly **60** includes a tamper **62** with a bottom portion **64**. It is to be considered that the wheeled axle **22** may be a support for the tamper **62** when inserting a bottom portion through the fastener base **44**. In a suitable embodiment, the wheeled axle **22** may be a support with

the fastener base **44** to the bottom portion **64** of the tamper **62** wherein the bottom portion **64** is enclosed by the wheeled axle **22** and the fastener base **44**.

(13) The foregoing description conveys the best understanding of the objectives and advantages of the present invention. Different embodiments may be made of the inventive concept of this invention. It is to be understood that all matter disclosed herein is to be interpreted merely as illustrative, and not in a limiting sense.

Claims

1. A wheel attachment for tamper comprising: a wheeled axle assembly including an axle with a box section body, said wheeled axle including bearings in each distal end thereof to allow to transport a tamper by wheels; a handle assembly including a handle, wherein said handle has a rectangular hollow body, said handle extends outwardly from said wheeled axle in a perpendicular configuration, said handle further being a driver for a tamper; a tamper assembly including said tamper, wherein said tamper has a bottom portion that is inserted through a fastener base, said tamper being lifted up by said handle; wherein said fastener base and said wheeled axle define a rectangular hollow portion into which the bottom portion of the tamper is inserted, and wherein the rectangular hollow portion has a configuration matching the tamper bottom portion to securely hold the tamper in position.
2. The wheel attachment for tamper of claim 1, wherein said wheeled axle includes bearings attached on the sides to permit said tamper to be transported.
3. The wheel attachment for tamper of claim 1, wherein said wheeled axle includes wheels attached on the sides thereof.
4. The wheel attachment for tamper of claim 3, wherein said wheels includes bearing portions concentric to said wheels configured to roll to transport said tamper.
5. The wheel attachment for the tamper of claim 1, wherein said wheeled axle and said fastener base define a rectangular hollow portion.
6. The wheel attachment for tamper of claim 1, wherein said wheeled axle is configured to support said tamper by tires wherein the tamper is inserted through said rectangular hollow portion defined by said fastener base and said wheeled axle.
7. The wheel attachment for tamper of claim 5, wherein at least one of the tires is attached to at least one of the wheels in cooperation with said handle which allow to transport said tamper.
8. The wheel attachment for tamper of claim 1, wherein said handle is attached to said wheeled axle in a center portion and has a rectangular hollow shape which is perpendicular to said fastener base both extending outwardly said wheeled axle.
9. The wheel attachment for tamper of claim 1, wherein said fastener extends from said wheeled axle by a pair of distal ends defining a rectangle configured to insert said bottom portion.
10. The wheel attachment for tamper of claim 1, wherein said fastener base is a support base for said bottom portion of said tamper.
11. A wheel attachment for tamper, comprising: a wheeled axle assembly including an axle with a box section body, wherein said wheeled axle includes bearings attached on each side thereof and at least one of the wheels is attached on one of said bearings, said wheels include tires wherein at least one of the tires is attached on at least one of the wheels; a handle assembly including a handle, wherein said handle includes a hollow rectangular body, said handle is attached on said wheeled axle, fastener base is perpendicularly attached to said handle, said wheeled axle and said fastener base define a rectangular hollow portion, said wheeled axle is configured to support said tamper by tires wherein the tamper is inserted through said rectangular hollow portion defined by said fastener base and said wheeled axle; and a tamper assembly including a tamper, wherein said tamper has a bottom portion, said bottom portion is attached to said fastener base and said handle; wherein the rectangular hollow portion has a configuration matching the tamper bottom portion to securely hold

the tamper in position.

12. The wheel attachment for tamper of claim 11, wherein at least one of the bearing portion is concentrically attached to at least one of the wheels to connect each side of the wheeled axle to said wheel through said bearings.

13. A wheel attachment for tamper, consisting of: a wheeled axle assembly including an axle, wherein said wheeled axle includes bearings attached on each side thereof and at least one of the wheels is attached on one of said bearings, said wheels include tires wherein at least one of the tires is attached on at least one of the wheels, at least one of said wheels has concentrically attached a bearing portion to connect each side of said wheeled axle, said wheeled axle configured to further transport a tamper by a fastener base and a handle; a handle assembly including said handle, wherein said handle has a hollow rectangular body which extends outwardly from said wheeled axle from a central portion, said handle is attached on said wheeled axle being perpendicular to a fastener base, said fastener base is perpendicularly attached to said handle and configured to insert a bottom portion of a tamper which is held by said wheeled axle said wheeled axle and said fastener base define a rectangular hollow portion, said wheeled axle is configured to support said tamper by tires wherein the tamper is inserted through said rectangular hollow portion defined by said fastener base and said wheeled axle; and a tamper assembly including said tamper mounted to said wheeled axle, said bottom portion is attached between said fastener base and said wheeled axle; wherein the rectangular hollow portion has a shape specifically designed to match the tamper bottom portion for secure retention during transportation.
